Practice Set 3: Wrong-Answer Notes

CFA Level II · 13 cards (11 wrong, 2 right but unsure), one per page

43 / 54

Practice set 4 (79.6%)

Score by topic

Topic	Score
Derivatives	3/5
Financial Statement Analysis	12/18
Equity Valuation	7/9
Fixed Income	7/8
Portfolio Management	5/5
Quantitative Methods	3/3
Corporate Issuers	2/2
Ethical and Professional Standards	4/4

Questions at a glance

Q	Topic	You	Ans
Q6	FSA	A	B
Q14	FSA	C	A
Q19	Deriv	C	A
Q21	FSA	B	A
Q24	FSA	C	B
Q28	Deriv	A	C
Q40	FSA	B	C
Q41	FSA	A	B
Q47	FI	C	A
Q49	Equity	A	B
Q50	Equity	C	C
Q53	Equity	B	C
Q54	Equity	A	A

Set 4 Q6 Bank funding and an inverted yield curve: banks borrow short and lend long

You **A** · Answer **B**

FSA

VIGNETTE (RELEVANT PART, UNEDITED)

Leah Morgan, a bank analyst, compares two banks. Exhibit 1 shows selected ratios for the last two years.

Bank X funds itself mainly with long-term consumer deposits; Bank Y relies on short-term interbank borrowing. Morgan expects the yield curve to invert next year.

Question 6. Which bank has the more stable funding, and what effect would an inverted yield curve most likely have on bank net interest revenue?

OPTIONS

A	Bank X; net interest revenue would rise	✗ Your pick	Your pick: Bank X is the right bank, but an inverted curve lowers, not raises, net interest revenue.
B	Bank X; net interest revenue would fall	✓ Correct	Correct: Bank X (consumer deposits are sticky) and net interest revenue falls.
C	Bank Y; net interest revenue would rise		Bank Y funds with short-term interbank borrowing, which is the LESS stable funding.

Trigger “Bank X funds itself mainly with long-term consumer deposits”
“Bank Y relies on short-term interbank borrowing”
“expects the yield curve to invert”

Rule next time: Stable funding = long-term, retail (consumer) deposits. An inverted curve (short rates above long rates) squeezes a bank that borrows short and lends long, so net interest revenue falls.

FORMULA / RULE

> Stable funding: retail and long-term deposits; unstable: short-term wholesale or interbank funding
>
> Banks: borrow short, lend long. Steeper curve → NIR up; flatter or inverted curve → NIR down

WHY THE ANSWER IS B

1. Bank X: long-term consumer deposits → more stable.
2. Inverted curve: short-term funding cost rises above long-term lending yield → NIR falls.
3. **B**.

Apply to other questions: Two parts, two checks: first pick the bank, then the direction. Do not answer the second part by instinct.

Set 4 Q14 **Partial goodwill: goodwill = price − your % × FV of net assets; NCI = their % × FV of net assets**

You **C** · Answer **A**

FSA

VIGNETTE (RELEVANT PART, UNEDITED)

Montclair Holdings acquires 80% of Tessaro SpA for €800 million in cash. The fair value of Tessaro's identifiable net assets is €850 million. Montclair's analyst compares the full goodwill and partial goodwill methods.

Question 14. Under the partial goodwill method, goodwill and the non-controlling interest are closest to:

OPTIONS

A	€120 million and €170 million	✓ Correct	Correct: goodwill €120m, NCI €170m.
B	€150 million and €200 million		Full goodwill method: implied 100% value = 800 ÷ 0.80 = 1,000; goodwill = 1,000 − 850 = 150; NCI = 20% × 1,000 = 200.
C	€120 million and €200 million	✗ Your pick	Your pick: goodwill €120m is partial, but NCI €200m is the full-method NCI. The two numbers must come from the same method.

Trigger “acquires 80% of Tessaro SpA for €800 million”
“fair value of Tessaro's identifiable net assets is €850 million”
“partial goodwill method”

Rule next time: Partial goodwill: the NCI is measured at its share of the FAIR VALUE OF NET ASSETS only (no goodwill in the NCI). Full goodwill: NCI at its fair value, which includes its share of goodwill.

FORMULA / RULE

Partial: goodwill = price paid − % acquired × FV of net assets; NCI = % not owned × FV of net assets

Full: implied value of 100% = price ÷ % acquired; goodwill = that value − FV of net assets; NCI = % not owned × that value

WHY THE ANSWER IS A

1. Goodwill = 800 − 0.80 × 850 = 800 − 680 = €120m.
2. NCI = 0.20 × 850 = €170m.
3. **A**.

Apply to other questions: Check the pair: both numbers come from the same method. If the NCI contains a goodwill part, the goodwill must be the full amount too.

Set 4 Q19 Option Greeks: for a long put delta and rho are negative; gamma and vega are positive

You **C** · Answer **A**

Deriv

VIGNETTE (RELEVANT PART, UNEDITED)

Ruth Okafor supervises the options desk at Tellis Securities. She reviews four positions: a long call, a short call, a long put and a short put, all on Ardent Mining shares. A client who owns 50,000 Ardent shares asks Okafor how buying put options on Ardent would change the risk of the client's position.

Question 19. Which Greeks are negative for a long put?

OPTIONS

A	Delta and rho	✓ Correct	Correct: delta (−) and rho (−) are negative for a long put.
B	Delta and gamma		Delta is negative, but gamma is positive for a long put.
C	Gamma and vega	✗ Your pick	Your pick: gamma and vega are the two Greeks that are POSITIVE for any long option.

Trigger “a long put” in the question: “Which Greeks are negative”
“long call, a short call, a long put and a short put”

Rule next time: Gamma and vega are positive for EVERY long option (call or put) and negative for every short option. Delta and rho have the sign of the option: negative for a put.

FORMULA / RULE

Long call: delta +, gamma +, vega +, theta −, rho +
Long put: delta −, gamma +, vega +, theta −, rho −
Short = flip every sign

WHY THE ANSWER IS A

1. Put value falls when the stock rises: delta −.
2. Put value falls when rates rise: rho −.
3. Gamma and vega are positive for every long option: **A**.

Apply to other questions: First sort the Greeks into the two groups: sign depends on call or put (delta, rho), or on long or short only (gamma, vega, theta).

Set 4 Q21 **RSU expense = grant-date fair value × RSUs vested, less the reversal for forfeitures**

You **B** · Answer **A**

FSA

VIGNETTE (RELEVANT PART, UNEDITED)

During the year 30,000 RSUs vested. They were granted at the start of the year with a grant-date fair value of $24 per RSU, and Kiruna's share price on the vesting date was $31. In the same year, 2,000 unvested RSUs from an earlier grant (grant-date fair value $22) were forfeited, and the full expense previously recognised for them ($22 per RSU) is reversed this year.

Question 21. Kiruna's RSU compensation expense for the year is closest to:

OPTIONS

A	$676,000	✓ Correct	Correct: 30,000 × 24 − 2,000 × 22 = $676,000.
B	$930,000	✗ Your pick	Your pick: $930,000 = 30,000 × $31 uses the vesting-date price, and it also ignores the forfeitures.
C	$720,000		$720,000 = 30,000 × $24 is right for the vested RSUs but forgets the reversal for the forfeitures.

Trigger “granted at the start of the year with a grant-date fair value of $24 per RSU”
“share price on the vesting date was $31”: only for tax
“the full expense previously recognised for them ($22 per RSU) is reversed”

Rule next time: Expense uses the GRANT-DATE fair value, never the vesting-date price. A forfeiture reverses the expense already recognised (at its own grant-date value).

FORMULA / RULE

RSU expense = vested × grant-date FV − forfeited × their grant-date FV

Vesting-date price → tax deduction (windfall or shortfall = (vesting price − grant FV) × RSUs × tax rate), not expense

WHY THE ANSWER IS A

1. Vested: 30,000 × $24 = $720,000.
2. Forfeited reversal: 2,000 × $22 = $44,000.
3. $720,000 − $44,000 = $676,000: **A**.

Apply to other questions: When two prices are given (grant and vesting), the expense takes the grant price and the tax takes the vesting price.

Set 4 Q24 **Vested RSUs are common shares (basic count); only unvested RSUs are potentially dilutive**

You **C** · Answer **B**

FSA

VIGNETTE (RELEVANT PART, UNEDITED)

During the year 30,000 RSUs vested. They were granted at the start of the year with a grant-date fair value of $24 per RSU, and Kiruna's share price on the vesting date was $31. In the same year, 2,000 unvested RSUs from an earlier grant (grant-date fair value $22) were forfeited, and the full expense previously recognised for them ($22 per RSU) is reversed this year.

Question 24. The vesting of the RSUs increases Kiruna's basic shares outstanding by:

OPTIONS

A	28,000 shares		28,000 subtracts the forfeited RSUs, but those were unvested RSUs from an earlier grant, never counted in basic shares.
B	30,000 shares	✓ Correct	Correct: 30,000 vested RSUs become 30,000 shares.
C	0 shares, because RSUs are only potentially dilutive	✗ Your pick	Your pick: 0 shares treats vested RSUs as still potentially dilutive. That is only true while they are unvested.

Trigger "30,000 RSUs vested"
"2,000 unvested RSUs … were forfeited"

Rule next time: When RSUs vest they turn into shares: basic shares rise by the number vested. Unvested RSUs are potentially dilutive (treasury stock method).

FORMULA / RULE

Basic shares ↑ by the number of RSUs that vest

Unvested RSUs: diluted shares only, using the treasury stock method (unrecognised expense counts as proceeds)

WHY THE ANSWER IS B

1. Vested RSUs → issued as common shares.
2. Basic shares rise by 30,000.
3. Forfeitures concern unvested RSUs and do not change the vested number: **B**.

Apply to other questions: Ask: vested or unvested? Vested → basic. Unvested → dilution test only.

Set 4 Q28 **Long FRA settled in advance: (MRR − FRA rate) × days/360 × notional, then discount at MRR**

You **A** · Answer **C**

Deriv

VIGNETTE (RELEVANT PART, UNEDITED)

Kaito Mori, a treasury analyst at Shirane Bank, entered a long 6 × 9 FRA (pay fixed, receive three-month MRR) on a notional of €25 million at an FRA rate of 2.40%. The FRA is settled in advance: the settlement amount is paid at FRA expiration, discounted at the three-month MRR.

Situation 2: At FRA expiration (month 6), three-month MRR is 2.85%.

Question 28. In Situation 2, the settlement amount Mori receives at FRA expiration is closest to:

OPTIONS

A	€28,125	✗ Your pick	Your pick: €28,125 is the interest difference at month 9 and is not yet discounted.
B	€112,500		€112,500 = 0.45% × 25,000,000 leaves out the 90/360 day fraction.
C	€27,926	✓ Correct	Correct: 28,125 ÷ (1 + 0.0285 × 90/360) = €27,926.

Trigger “The FRA is settled in advance: the settlement amount is paid at FRA expiration, discounted at the three-month MRR”
“At FRA expiration (month 6), three-month MRR is 2.85%”

Rule next time: The interest difference belongs to month 9, but the FRA settles at month 6, so you must discount it back 90 days at the then-current MRR.

FORMULA / RULE

Payment (long FRA) = (MRR − FRA rate) × (days/360) × notional ÷ (1 + MRR × days/360)
Long FRA = pay fixed: gains when MRR rises above the FRA rate

WHY THE ANSWER IS C

1. Interest = (0.0285 − 0.0240) × 90/360 × 25,000,000 = €28,125.
2. Discount: 1 + 0.0285 × 0.25 = 1.007125.
3. 28,125 ÷ 1.007125 = €27,926: **C**.

Apply to other questions: “Settled in advance” or “paid at expiration” in a FRA question means a final discounting step.

Set 4 Q40 Remeasurement and translation: net monetary liabilities lose when the foreign currency rises; net assets gain

You **B** · Answer **C**

FSA

VIGNETTE (RELEVANT PART, UNEDITED)

Ruiz compares two foreign subsidiaries of Atlas. Both have net monetary liabilities and positive net assets.

Subsidiary A's functional currency is the US dollar (the parent's presentation currency), so it is remeasured with the temporal method.

Subsidiary B's functional currency is its local currency, so it is translated with the current rate method.

During the year both local currencies appreciate against the dollar. Prices in both countries are rising, and Subsidiary B uses FIFO.

Question 40. Because the local currency appreciated, Subsidiary A's remeasurement result and Subsidiary B's CTA are most likely:

OPTIONS

A	a loss for both		Not a loss for both: with positive net assets and a rising foreign currency, the current rate method gives a positive CTA, not a loss.
B	a gain in income for A and a negative CTA for B	✗ Your pick	Your pick: this is the result for a FALLING foreign currency (liabilities worth less → gain; net assets worth less → negative CTA). The direction is reversed.
C	a loss in income for A and a positive CTA for B	✓ Correct	Correct: loss in income for A and a positive CTA for B.

Trigger “both local currencies appreciate against the dollar” (local currency = the subsidiary's own, foreign, currency)
“net monetary liabilities and positive net assets”
“Subsidiary A … temporal method”; “Subsidiary B … current rate method”

Rule next time: “Local currency” is the subsidiary's currency, i.e. the FOREIGN currency. Local currency appreciates = foreign currency up. Net monetary liabilities (temporal) then cost more in dollars → loss. Net assets (current rate) are worth more in dollars → positive CTA.

FORMULA / RULE

Temporal: exposed item = net monetary assets (liabilities if net liabilities); FC ↑ → net monetary asset gain, net monetary liability LOSS (goes to net income)

Current rate: exposed item = net assets; FC ↑ → positive CTA, FC ↓ → negative CTA (goes to equity)

WHY THE ANSWER IS C

1. Local currency appreciates → foreign currency ↑ vs the parent's currency.
2. A: net monetary liabilities now cost more dollars → remeasurement loss in net income.
3. B: positive net assets worth more dollars → CTA ↑ (positive): **C**.

Apply to other questions: Translate the words first: “local currency up” = foreign currency up. Then use the table for FC ↑. The “FC ↓” row (retained earnings ↑, CTA ↓ for a net-liability-in-temporal and net-asset-in-current-rate pair) is the opposite case.

Set 4 Q41 **Temporal method with a rising foreign currency and rising prices: FIFO gives the highest gross margin**

You **A** · Answer **B**

FSA

VIGNETTE (RELEVANT PART, UNEDITED)

During the year both local currencies appreciate against the dollar. Prices in both countries are rising, and Subsidiary B uses FIFO.

Question 41. For a subsidiary with an appreciating local currency and rising prices, which choices give the higher translated gross margin?

OPTIONS

A	Temporal method and LIFO	✗ Your pick	Your pick: temporal is right, but LIFO in rising prices puts the newest, highest costs into COGS and lowers the margin.
B	Temporal method and FIFO	✓ Correct	Correct: temporal method and FIFO.
C	Current rate method and LIFO		Current rate method translates COGS at the average rate as well, so there is no currency boost to the margin; with LIFO the margin is lower still.

Trigger “appreciating local currency” in the question
“rising prices”

Rule next time: Temporal: COGS is translated at old (lower) historical rates while sales use the average rate, so margin rises when the foreign currency appreciates. In rising prices FIFO puts the older, cheaper costs into COGS.

FORMULA / RULE

> FC ↑: temporal gives the higher gross margin; FC ↓: current rate gives the higher gross margin
> Rising prices: FIFO → higher margin; falling prices: LIFO → higher margin

WHY THE ANSWER IS B

1. Temporal method: COGS at older, lower historical rates → margin ↑ when FC rises.
2. FIFO in rising prices: older, cheaper costs in COGS → margin ↑.
3. Both → **B**.

Apply to other questions: Two independent switches: currency direction picks the method, price direction picks the inventory method.

Set 4 Q47 Capped floater on a binomial tree: cap the coupon at each node, then work backward

You **C** · Answer **A**

FI

VIGNETTE (RELEVANT PART, UNEDITED)

Silva values a two-year floating-rate note with an annual coupon equal to the one-year reference rate set at the start of each year, paid at the end of the year. The coupon is capped at 4.8%. Exhibit 1 shows the calibrated binomial interest rate tree (equal probabilities).

Exhibit 1: One-Year Rates

Node	Rate
Time 0	3.20%
Time 1, upper node	5.40%
Time 1, lower node	3.80%

Question 47. The value of the capped floater is closest to:

OPTIONS

A	99.72	✓ Correct	Correct: 99.72.
B	99.43		99.43 is only the value at the upper time-1 node, not the value today.
C	100.00	✗ Your pick	Your pick: 100.00 is the value of the UNCAPPED floater. The cap lowers the value.

Trigger “coupon is capped at 4.8%”
“coupon equal to the one-year reference rate set at the start of each year, paid at the end of the year”

Rule next time: Rate 5.40% is above the 4.80% cap, so the upper node pays 4.80 only. The coupon is set at the start and paid at the end, so the node's own rate is both the coupon (capped) and the discount rate.

FORMULA / RULE

> Node value = (next values + coupon) ÷ (1 + node rate); coupon = min(node rate, cap)
> Time-0 value = average of the two time-1 values plus the time-0 coupon, ÷ (1 + r0)

WHY THE ANSWER IS A

1. Upper: (100 + 4.80) ÷ 1.054 = 99.4307.
2. Lower: (100 + 3.80) ÷ 1.038 = 100.0000.
3. Time 0: [0.5 × (99.4307 + 3.20) + 0.5 × (100.0000 + 3.20)] ÷ 1.032 = 99.72: **A**.

Apply to other questions: Cap on a floater: check each node's rate against the cap before discounting.

Set 4 Q49 **Residual income = net income − equity charge (cost of equity × beginning equity)**

You **A** · Answer **B**

Equity

VIGNETTE (RELEVANT PART, UNEDITED)

Orchard Packaging Group (OPG) had total assets of $2,400,000, financed with three times as much debt capital as equity capital.

OPG's pretax cost of debt is 7% and its cost of equity capital is 11%.

OPG had EBIT of $280,000 and was taxed at a rate of 30%. EBIT is expected to continue at $280,000 indefinitely.

Question 49. Using the method that deducts an equity charge, OPG's residual income is closest to:

OPTIONS

A	$107,800.	✗ Your pick	Your pick: $107,800 is the net income. The equity charge of $66,000 still has to be deducted.
B	$41,800.	✓ Correct	Correct: 107,800 − 66,000 = $41,800.
C	$130,000.		$130,000 = EBIT × (1 − 30%) − 66,000 ignores the interest on the debt, so it is not net income.

Trigger “the method that deducts an equity charge”
“three times as much debt capital as equity capital”
“pretax cost of debt is 7%” and “cost of equity capital is 11%”

Rule next time: “Deducts an equity charge” means RI = net income − cost of equity × equity. Net income is after interest and tax, so first work out the debt and the interest.

FORMULA / RULE

RI = NI − r × beginning book value of equity

NI = (EBIT − interest) × (1 − t)

3 times as much debt as equity → debt = 3/4, equity = 1/4 of total assets

WHY THE ANSWER IS B

1. Equity = 600,000, debt = 1,800,000; interest = 126,000.
2. NI = (280,000 − 126,000) × 0.70 = 107,800.
3. RI = 107,800 − 0.11 × 600,000 = 41,800: **B**.

Apply to other questions: EVA-style (NOPAT − WACC × capital) and RI (NI − Ke × equity) are two routes: use the one the question names.

Set 4 Q50 **Perpetual residual income: value = book value per share + RI per share ÷ r**

You **C** · Answer **C**

Equity

VIGNETTE (RELEVANT PART, UNEDITED)

Orchard Packaging Group (OPG) had total assets of $2,400,000, financed with three times as much debt capital as equity capital.

OPG's book value per share is $15 and it has 40,000 shares of common stock outstanding.

OPG had EBIT of $280,000 and was taxed at a rate of 30%. EBIT is expected to continue at $280,000 indefinitely.

OPG's pretax cost of debt is 7% and its cost of equity capital is 11%.

Question 50. Using the residual income model, with the residual income expected to continue indefinitely, OPG's intrinsic value per share is closest to:

OPTIONS

A	$9.50.		$9.50 is only the PV of the residual income; the book value is missing.
B	$16.05.		$16.05 = 15.00 + 1.045 counts one year of RI instead of a perpetuity.
C	$24.50.	✓ Correct	Correct (your pick): 15.00 + 1.045 ÷ 0.11 = $24.50.

Trigger “EBIT is expected to continue at $280,000 indefinitely”
“residual income expected to continue indefinitely”
“book value per share is $15 and it has 40,000 shares”

Rule next time: Indefinite constant RI is a perpetuity: PV = RI ÷ r. Intrinsic value is book value PLUS that PV.

FORMULA / RULE

V0 = B0 + RI1 ÷ r (RI constant forever)
RI per share = total RI ÷ shares

WHY THE ANSWER IS C

1. RI per share = 41,800 ÷ 40,000 = $1.045.
2. PV = 1.045 ÷ 0.11 = $9.50.
3. V0 = 15.00 + 9.50 = $24.50: **C**.

Apply to other questions: “Indefinitely” → divide by r. “For one year only” → divide by (1 + r).

Set 4 Q53 **Residual income recognises value EARLIER; in theory it gives the same value as DDM and FCF**

You **B** · Answer **C**

Equity

VIGNETTE (RELEVANT PART, UNEDITED)

Stark asks Parker to compare residual income models to dividend discount models and free cash flow models. Parker recognizes that valuation models based on discounting dividends or discounting free cash flows are as theoretically sound as the residual income model. While investigating the relationship of the residual income model to other valuation models, Parker makes the following statements.

Statement 1: Residual income valuations recognize value later than different present value approaches.

Statement 2: In theory, residual income valuations are mutually consistent with other present value approaches.

Question 53. Are Parker's statements comparing residual income models with different present value approaches both correct?

OPTIONS

A	Yes		Both statements are not correct: Statement 1 is wrong.
B	No, only statement 1 is correct	✗ Your pick	Your pick: it is Statement 1 that is wrong, not Statement 2. RI recognises value earlier, not later.
C	No, only statement 2 is correct	✓ Correct	Correct: only Statement 2 is correct.

Trigger “Statement 1: Residual income valuations recognize value later than different present value approaches”
“Statement 2: In theory, residual income valuations are mutually consistent with other present value approaches”

Rule next time: The book value is already in the starting value, so RI recognises value earlier (a larger part of the value sits in book value). Under the same assumptions all the models give the same value.

FORMULA / RULE

RI: value = B0 + PV(RI), so book value is recognised at once: earlier than DDM or FCF
Theory: DDM, FCFE and RI agree if the inputs are consistent (clean surplus)

WHY THE ANSWER IS C

1. Statement 1: wrong (earlier, not later).
2. Statement 2: right (consistent in theory).
3. **C**.

Apply to other questions: Memorise the pair: RI = earlier recognition of value + same value in theory.

Set 4 Q54 **Two-stage residual income with a P/B terminal value**

You **A** · Answer **A**

Equity

VIGNETTE (RELEVANT PART, UNEDITED)

Stark asks Parker to evaluate whether the common stock of Company D is undervalued, fairly valued, or overvalued based on a residual income model. Parker uses the two-stage residual income model to value Company D's share. For her analysis, she assumes the following:

Current market price is $24.00.

Beginning book value per share is $20.00.

Earnings per share will be 32% of beginning book value for the next year.

Cash dividends will be 50% of earnings per share for the next year.

At the end of the year, the stock's P/B ratio will be 1.1.

Required rate of return on equity is 10%.

Question 54. Based on the information given, Parker should conclude that Company D's share is:

OPTIONS

A	undervalued.	✓ Correct	Correct (your pick): value $26.11 is above the price of $24: undervalued.
B	fairly valued.		Fairly valued would need value = $24.
C	overvalued.		Overvalued would need value below $24.

Trigger "Earnings per share will be 32% of beginning book value"
"Cash dividends will be 50% of earnings per share"
"the stock's P/B ratio will be 1.1"
Rule next time: Year 1: RI = EPS − r × B0. Terminal value at year-end = (P/B − 1) × ending book value (the excess over book). Add B0 + PV(RI1) + PV(terminal premium) and compare with the price.

FORMULA / RULE

B1 = B0 + EPS − DPS
RI1 = EPS1 − r × B0
V0 = B0 + RI1 ÷ (1 + r) + (P/B − 1) × B1 ÷ (1 + r)

WHY THE ANSWER IS A

1. EPS = 0.32 × 20 = 6.40; DPS = 3.20; B1 = 20 + 6.40 − 3.20 = 23.20.
2. RI1 = 6.40 − 0.10 × 20 = 4.40; PV = 4.00.
3. Terminal premium = 0.1 × 23.20 = 2.32; PV = 2.11. V0 = 20 + 4.00 + 2.11 = $26.11 > $24: **A**.

Apply to other questions: A terminal P/B of 1.1 means a terminal RI value of (1.1 − 1) × B1, not the whole price.